

The effect of different sodium hydroxide concentrations on CO₂ absorption in a system with *Chlorella vulgaris* encapsulated in calcium alginate beads

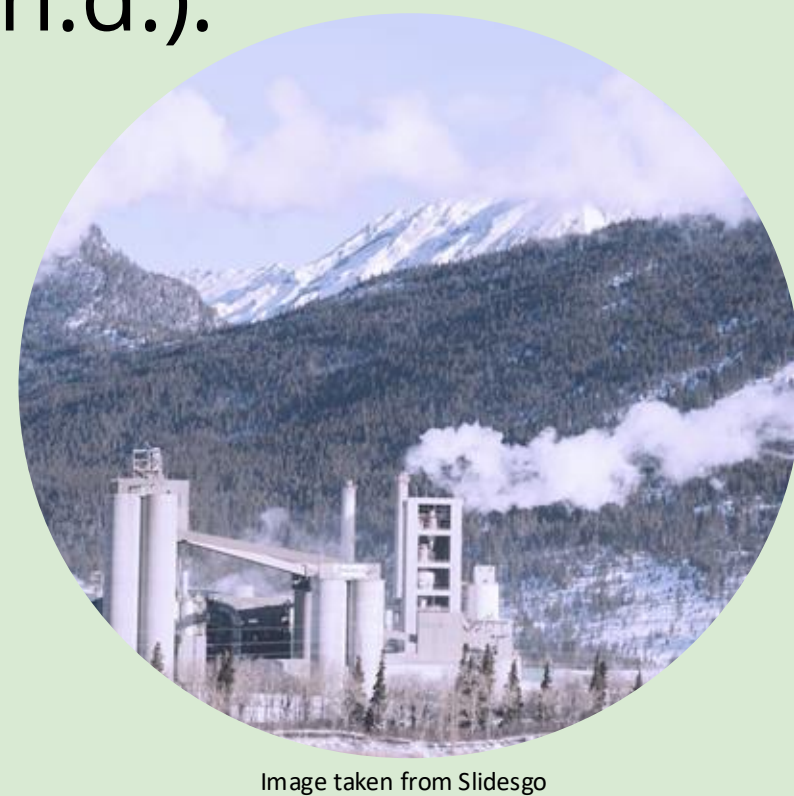
Environmental Engineering

Introduction

Carbon Dioxide (CO₂) emissions drive **climate change** by trapping heat in the atmosphere through the **greenhouse effect**. This causes **rising temperatures**, ocean **acidification**, and changing weather pattern, leading scientists to explore ways to reduce CO₂ levels.

One approach is **chemical absorption** using sodium hydroxide (**NaOH**), which forms **carbonate compounds**. Another method is **biological fixation** using *Chlorella vulgaris* through photosynthesis and converts it into biomass (Sadewo et al., 2022).

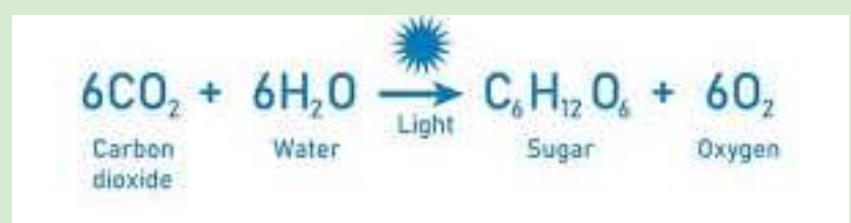
High NaOH concentrations can harm microalgae, but calcium **alginate encapsulation** may **protect** the **cells** while still allowing CO₂ diffusion (Damayanti et al., n.d.).



Overall Chemical Reaction



Photosynthesis Equation



Purpose

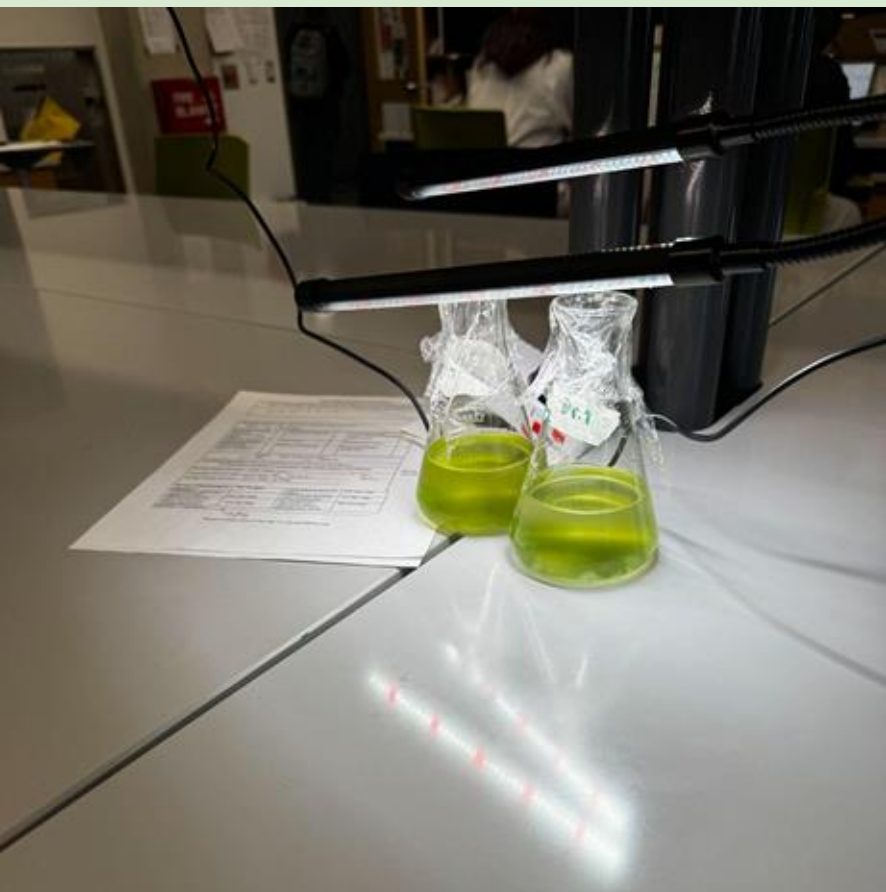
This project explores whether **combining chemical** absorption using NaOH with **biological** carbon fixation by microalgae can improve overall CO₂ capture **efficiency**. Specifically, it investigates how **different** initial NaOH **concentrations** affect CO₂ absorption when NaOH is introduced gradually into a system containing immobilized microalgae. Additionally, this study examines whether alginate encapsulation enables **microalgae** to remain **viable** and function effectively **under alkaline conditions**.

Methods

PHASE 1

Subculture the Microalgae:

Grow the *Chlorella vulgaris* culture in BG-11 growth medium for about a month. During this period, the absorption of the algae is collected using a spectrophotometer every class period.



PHASE 2

Encapsulation of Microalgae:

Mix the microalgae with a carboxymethylcellulose & calcium lactate solution to create a stock solution. Next slowly drip the stock solution into a sodium alginate bath to create the beads.



Independent Variable:

Initial concentration of the sodium hydroxide introduced to the system (0.2, 0.4, 0.6, 0.8, 1 M)

Dependent Variable:

Amount of CO₂ absorbed by the system

Element of Design

Hypothesis

It is hypothesized that moderate initial concentrations of sodium hydroxide will maximize CO₂ absorption

Alginate Beads with Microalgae, no NaOH

Microalgae and NaOH, no alginate bead

NaOH only system

PHASE 3

Experimental System Setup:

Place the alginate beads that contain the *Chlorella vulgaris* into the filter flask. Prepare the equipment needed for controlled chemical introduction and gas exposure.



PHASE 4

CO₂ Exposure and NaOH Introduction

Introduce carbon dioxide into the system while simultaneously adding the sodium hydroxide solution using a controlled dispensing system. This allows for the CO₂ to interact with both the chemical absorbent and the encapsulated microalgae.



Results

At this stage, the experimental design and **preparation** are **complete**. *Chlorella vulgaris* **cultures** have been successfully **grown** and standardized, and methods for encapsulating the cells in calcium alginate beads have been established. The **setup** was **assembled** and **tested**, but the **flow meter** measurements were **oscillating** due to the pump system **variations**. This **prevented data** collection for any control **test** runs or for any experimental tests.

Sodium hydroxide is **expected** to **chemically absorb** CO₂ through reactions that form carbonate compounds, while *Chlorella vulgaris* will **biologically absorb** CO₂ through photosynthesis. The alginate **beads** should **protect** the microalgae from high pH conditions while still allowing gas exchange (Tran et al., 2024). **Moderate** NaOH **concentrations** are expected to result in the **highest** overall CO₂ **absorption** by balancing strong chemical capture with biological activity.

Conclusions

If **expected** results are **observed**, they would suggest that combining chemical and biological carbon capture methods can **improve** overall **efficiency**. NaOH would provide rapid chemical absorption, while *Chlorella vulgaris* would contribute sustained CO₂ uptake through photosynthesis. The alginate beads may play a key role by protecting the microalgae from harmful pH conditions. Identifying an **optimal** NaOH concentration **range** could help improve the **design** of **hybrid** carbon capture **systems** for more stable and effective **real-world applications**.

Extensions

A possible extension of this research is **adapting** the system so it can be used in **real-world** carbon capture **applications**. For example, a larger reactor or filtration system could use the hybrid system to capture CO₂ from **industrial emissions** or **vehicle exhaust**.

Future research could focus on **optimizing** bead **size**, algae **concentration**, and reactor **design** to improve CO₂ capture efficiency and determine whether this system is practical for reducing greenhouse gas emissions.

References

Damayanti, A., Kumoro, A. C., & Bahlawan, Z. A. S. (n.d.). Review Calcium Alginate Beads as Immobilizing Matrix of Functional Cells: Extrusion Dripping Method, Characteristics, and Application. *IOP Science*. 10.1088/1757-899X/1053/1/012017

Sadewo, R. P., Hidayati, N., Ambarsari, L., & Anam, K. (2022, November 11). CO₂ Sequestration Using Sodium Hydroxide and Its Utilization for Chlorella sorokiniana Biomass Production. *Research Gate*, 14(3), 391-399.

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